WORKED SOLUTION

STEP I 2005, Question 9

Mechanics • Rigid Body Equilibrium

> **STEP I 2005, Question 9**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/05-s1-q9

The key idea

Everything hinges on one unknown: where the centre of mass G of the non-uniform rod lies. The first two set-ups each give T in terms of that position, so equating them pins down G and T together. After that, the tilted set-up is a standard rigid-body problem: take moments about the point where the unknown contact forces act, then resolve.

Solution

We model the rod as rigid, with all its weight W acting at its centre of mass G, and the strings as light and inextensible. Let $AG = x$, so $GB = 3l - x$.

Finding G and showing $5T = 2W$

Strings at both ends. Taking moments about B removes the unknown tension at B:

$$T \cdot 3l = W(3l - x).$$

Pivot at distance l from A, string at B. Let P be the pivot, so $PB = 2l$. Taking moments about P removes the unknown pivot reaction. For the string at B to hold the rod up, G must lie on the B side of P, so $PG = x - l$ and

$$T \cdot 2l = W(x - l).$$

Eliminating T (multiply the first equation by 2, the second by 3, and compare):

$$2W(3l - x) = 3W(x - l) \quad \Longrightarrow \quad 9l = 5x \quad \Longrightarrow \quad x = \tfrac{9}{5}l.$$

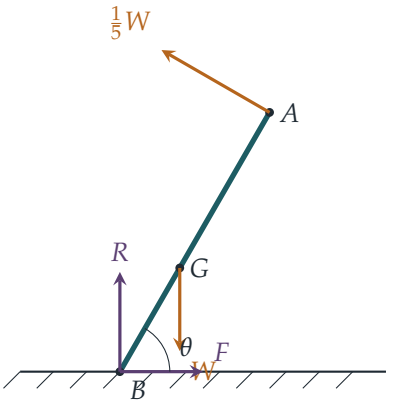
Then, from the second equation, $2lT = W \cdot \frac{4}{5}l$, so $T = \frac{2}{5}W$, that is,

$$5T = 2W,$$

as required. Note that $x = \frac{9}{5}l > l$, which agrees with our assumption that G is beyond the pivot. So G is at distance $GB = 3l - \frac{9}{5}l = \frac{6}{5}l$ from B.

The rod resting on the ground

Now B is on rough ground, the rod makes angle θ with the horizontal, and the string at A is perpendicular to the rod with tension $\frac{1}{2}T = \frac{1}{5}W$. At B there is a normal reaction R upwards and a friction force F along the ground.



Moments about B. This kills R and F. The string force is perpendicular to the rod at distance $3l$ from B, so its moment is $\frac{1}{5}W \cdot 3l$. The weight acts vertically at distance $\frac{6}{5}l$ along the rod, so its perpendicular distance from B is $\frac{6}{5}l\cos\theta$. Hence

$$\tfrac{1}{5}W \cdot 3l = W \cdot \tfrac{6}{5}l\cos\theta \quad\Longrightarrow\quad \cos\theta = \tfrac{1}{2},$$

so $\theta = 60^\circ$ (that is, $\frac{\pi}{3}$).

Ask yourself Which way do the components of the string's tension point?

The rod makes angle θ with the horizontal, and the string is at right angles to the rod, so the string makes angle θ with the *vertical*. Its tension $\frac{1}{5}W$ therefore has vertical component $\frac{1}{5}W\cos\theta$ (upwards) and horizontal component $\frac{1}{5}W\sin\theta$, pointing back over B, away from the side of A. Friction must balance this horizontal pull, so it acts at B towards the side of A, as drawn.

Resolving vertically:

$$R + \tfrac{1}{5}W\cos 60^\circ = W \quad\Longrightarrow\quad R = W - \tfrac{1}{10}W = \tfrac{9}{10}W.$$

Resolving horizontally:

$$F = \tfrac{1}{5}W\sin 60^\circ = \tfrac{\sqrt{3}}{10}W.$$

The rod does not slip provided $F \le \mu R$, that is,

$$\mu \ge \frac{F}{R} = \frac{\sqrt{3}/10}{9/10} = \frac{\sqrt{3}}{9}.$$

So the smallest coefficient of friction that prevents slipping is $\mu = \dfrac{\sqrt{3}}{9} = \dfrac{1}{3\sqrt{3}} \approx 0.192$ (with equality the rod is in limiting equilibrium).

Checks

- With $x = \frac{9}{5}l$, the two-string set-up gives $T = W \cdot \frac{6/5}{3} = \frac{2}{5}W$, and the pivot set-up gives $T = W \cdot \frac{4/5}{2} = \frac{2}{5}W$: the same, as it must be.
- In the tilted position, taking moments about A as well (with $W = l = 1$, $\theta = 60°$, $R = 0.9$, $F \approx 0.1732$) also gives zero resultant moment, a check on R and F together.
- Numerically $\mu_{\min} = \sqrt{3}/9 \approx 0.19245$, which a computer confirmed by building the force vectors and checking all three equilibrium equations.

Common slips

- Using $\frac{6}{5}l\sin\theta$ for the weight's moment arm. The horizontal distance from B to the line of the weight is $\frac{6}{5}l\cos\theta$.
- Forgetting that the perpendicular string has a vertical component, and taking $R = W$. That gives $\mu = \sqrt{3}/10$, which is too small.
- Assuming the rod is uniform, so that G is at the midpoint. The whole point of the first part is to locate G.

Beyond STEP

Locating a centre of mass from equilibrium measurements is the static version of a powerful idea: the moment of a force system about a point determines the system, up to a single resultant. At university this becomes the study of rigid-body statics and then dynamics, where the centre of mass and the moment of inertia are the two numbers that govern a body's motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats torque and rigid bodies with a wealth of harder problems in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.